Controller Trade Study: PID vs. LQR

Criterion	Proportional-Integral Derivative (PID)	Linear-Quadratic Regulator (LQR)
Development	Tune Feedback loops in stages	Develop state-feedback gains and state estimates
Coupled motion	Check Interactions between feedback loops	Represent Coupling in the state-space model
Motor Limits	Check Saturation and anti-windup	Check saturation despite control-effort penalties

- Proposed baseline: PID for initial hover development
- PID allows Incremental tuning that will make early testing easier.
- LQR will remain an alternative just in case.
- Comparison: test PID and LQR under the same simulated conditions and compare the following: tracking accuracy, response time, overshoot, control effort, and sensitivity to parameter changes.

S&C Design Decisions for Discussion

Decision	My Proposed Direction	Needed Team Input
Initial Drone Model	Provisional Eight-rotor layout for hover	Rotor positions and Spin from Propulsion and Structures
Flight Software	Confirm existing ArduPilot tuning or a custom controller	Software scope and interfaces from Electronics (and myself)
Control Margins	Evaluate CG shifts and rotor thrust limits	Mass, inertia, and thrust from Structures and Propulsion
Titan Sea Operations	Define touchdown limits and takeoff conditions after sampling (if)	Landing limits and sample mass from Structures (and whoever is working on the flotation device)

- Next S&C deliverable:

Preliminary control margins and acceptance limits for the PDR presentation.